

2.3 Detailed Description of Phase II

In this section, we summarize the different steps of the simplex method we have described in the previous section. In fact, what we have described so far constitutes Phase II of the simplex method. Phase I deals with the problem of putting the linear program in the required form. This will be described in a later section.

Phase II of the simplex method

1. Suppose the initial or current tableau is

$-z$	x_1	$\dots$	x_s	$\dots$	x_n	
$\bar{c}_1$	$\dots$	$\bar{c}_s$	$\dots$	$\bar{c}_n$	$-\bar{c}_0$	
$\bar{a}_{11}$	$\dots$	$\bar{a}_{1s}$	$\dots$	$\bar{a}_{1n}$	$\bar{b}_1 \geq 0$	
$\vdots$		$\vdots$		$\vdots$	$\vdots$	
$\bar{a}_{r1}$	$\dots$	$\bar{a}_{rs}$	$\dots$	$\bar{a}_{rn}$	$\bar{b}_r \geq 0$	
$\vdots$		$\vdots$		$\vdots$	$\vdots$	
$\bar{a}_{m1}$	$\dots$	$\bar{a}_{ms}$	$\dots$	$\bar{a}_{mn}$	$\bar{b}_m \geq 0$	

and the variables can be partitioned into $B = \{x_{j_1}, \dots, x_{j_m}\}$ and N with

- $\bar{c}_{j_i} = 0$ for $i = 1, \dots, m$ and
-

$$\bar{a}_{kj_i} = \begin{cases} 0 & k \neq i \\ 1 & k = i. \end{cases}$$

The current basic feasible solution is given by $x_{j_i} = \bar{b}_i$ for $i = 1, \dots, m$ and $x_j = 0$ otherwise. The objective function value of this solution is $\bar{c}_0$.

2. If $\bar{c}_j \leq 0$ for all $j = 1, \dots, n$ then the current basic feasible solution is optimal. STOP.
3. Find a column s for which $\bar{c}_s > 0$. x_s is the variable entering the basis.
4. Check for unboundedness. If $\bar{a}_{is} \leq 0$ for $i = 1, \dots, m$ then the linear program is unbounded. STOP.
5. Min ratio test. Find row r such that

$$\frac{\bar{b}_r}{\bar{a}_{rs}} = \min_{i: \bar{a}_{is} > 0} \frac{\bar{b}_i}{\bar{a}_{is}}.$$

6. Pivot on $\bar{a}_{rs}$. I.e. replace the current tableau by:

	...	x_s	...	x_j	...	
$-z$	...	0	...	$\bar{c}_j - \frac{\bar{a}_{rj}\bar{c}_s}{\bar{a}_{rs}}$	...	$-\bar{c}_0 - \frac{\bar{b}_r\bar{c}_s}{\bar{a}_{rs}}$
		$\vdots$		$\vdots$		$\vdots$
row r	...	1	...	$\frac{\bar{a}_{rj}}{\bar{a}_{rs}}$	...	$\frac{\bar{b}_r}{\bar{a}_{rs}}$
		$\vdots$		$\vdots$		$\vdots$
row i	...	0	...	$\bar{a}_{ij} - \frac{\bar{a}_{rj}\bar{a}_{is}}{\bar{a}_{rs}}$	...	$\bar{b}_i - \frac{\bar{b}_r\bar{a}_{is}}{\bar{a}_{rs}}$
		$\vdots$		$\vdots$		$\vdots$

Replace x_{j_r} by x_s in B .

7. Go to step 2.

2.4 Convergence of the Simplex Method

As we have seen, the simplex method is an iterative method that generates a sequence of basic feasible solutions. But, do we have any guarantee that this process eventually terminates? The answer is yes if the linear program is nondegenerate.

Theorem 2.1 *The simplex method solves a nondegenerate linear program in finitely many iterations.*

Proof:

For nondegenerate linear programs, we have a strict improvement (namely of value $\frac{\bar{b}_r\bar{c}_s}{\bar{a}_{rs}} > 0$) in the objective function value at each iteration. This means that, in the sequence of bfs produced by the simplex method, each bfs can appear at most once. Therefore, for nondegenerate linear programs, the number of iterations is certainly upper bounded by the number of bfs. This latter number is finite (for example, it is upper bounded by $\binom{n}{m}$) since any bfs corresponds to m variables being basic³. QED.

However, when the linear program is degenerate, we might have degenerate pivots which give no strict improvement in the objective function. As a result, a subsequence of bases might repeat implying the nontermination of the method. This phenomenon is called *cycling*.

³Not all choices of basic variables give rise to *feasible* solutions.

2.4.1 An Example of Cycling

The following is an example that will cycle if unfortunate choices of entering and leaving variables are made (the pivot element is within a box).

$-z$	x_1	x_2	x_3	x_4	x_5	x_6	
	4	1.92	-16	-0.96			0
	-12.5	-2	12.5	1	1		0
	1	0.24	-2	-0.24		1	0

$-z$	x_1	x_2	x_3	x_4	x_5	x_6	
		0.96	-8	0		-4	0
		1	-12.5	-2	1	12.5	0
	1	0.24	-2	-0.24		1	0

$-z$	x_1	x_2	x_3	x_4	x_5	x_6	
			4	1.92	-0.96	-16	0
		1	-12.5	-2	1	12.5	0
	1		1	0.24	-0.24	-2	0

$-z$	x_1	x_2	x_3	x_4	x_5	x_6	
	-4			0.96	0	-8	0
	12.5	1		1	-2	-12.5	0
	1		1	0.24	-0.24	-2	0

$-z$	x_1	x_2	x_3	x_4	x_5	x_6	
	-16	-0.96			1.92	4	0
	12.5	1		1	-2	-12.5	0
	-2	-0.24	1		0.24	1	0

$-z$	x_1	x_2	x_3	x_4	x_5	x_6	
	-8	0	-4		0.96		0
	-12.5	-2	12.5	1	1		0
	-2	-0.24	1		0.24	1	0

	x_1	x_2	x_3	x_4	x_5	x_6	
$-z$	4	1.92	-16	-0.96			0
	-12.5	-2	12.5	1	1		0
	1	0.24	-2	-0.24		1	0

2.4.2 Bland's Anticycling Rule

The simplex method, as described in the previous section, is ambiguous. First, if we have several variables with a positive $\bar{c}_s$ (cfr. Step 3) we have not specified which will enter the basis. Moreover, there might be several variables attaining the minimum in the minimum ratio test (Step 5). If so, we need to specify which of these variables will leave the basis. A *pivoting rule* consists of an *entering variable rule* and a *leaving variable rule* that unambiguously decide what will be the entering and leaving variables.

The most classical entering variable rule is:

Largest coefficient entering variable rule: Select the variable x_s with the largest $\bar{c}_s > 0$. In case of ties, select the one with the smallest subscript s .

The corresponding leaving variable rule is:

Largest coefficient leaving variable rule: Among all rows attaining the minimum in the minimum ratio test, select the one with the largest pivot $\bar{a}_{rs}$. In case of ties, select the one with the smallest subscript r .

The example of subsection 2.4.1 shows that the use of the largest coefficient entering and leaving variable rules does not prevent cycling. There are two rules that avoid cycling: the lexicographic rule and Bland's rule (after R. Bland who discovered it in 1976). We'll just describe the latter one, which is conceptually the simplest.

Bland's anticycling pivoting rule: Among all variables x_s with positive $\bar{c}_s$, select the one with the smallest subscript s . Among the eligible (according to the minimum ratio test) leaving variables x_l , select the one with the smallest subscript l .

Theorem 2.2 *The simplex method with Bland's anticycling pivoting rule terminates after a finite number of iterations.*

Proof:

The proof is by contradiction. If the method does not stop after a finite number of iterations then there is a cycle of tableaus that repeats. If we delete from the tableau that initiates this cycle the rows and columns not containing pivots during the cycle, the

resulting tableau has a cycle with the same pivots. For this tableau, all right-hand-sides are zero throughout the cycle since all pivots are degenerate.

Let t be the largest subscript of the variables remaining. Consider the tableau T_1 in the cycle with x_t leaving. Let $B = \{x_{j_1}, \dots, x_{j_m}\}$ be the corresponding basis (say $j_r = t$), x_s be the associated entering variable and, a_{ij}^1 and c_j^1 the constraint and cost coefficients. On the other hand, consider the tableau T_2 with x_t entering and denotes by a_{ij}^2 and c_j^2 the corresponding constraint and cost coefficients.

Let x be the (infeasible) solution obtained by letting the nonbasic variables in T_1 be zero except for $x_s = -1$. Since all RHS are zero, we deduce that $x_{j_i} = a_{is}$ for $i = 1, \dots, m$. Since T_2 is obtained from T_1 by elementary row operations, x must have the same objective function value in T_1 and T_2 . This means that

$$c_0^1 - c_s^1 = c_0^2 - c_s^2 + \sum_{i=1}^m a_{is}^1 c_{j_i}^2.$$

Since we have no improvement in objective function in the cycle, we have $c_0^1 = c_0^2$. Moreover, $c_s^1 > 0$ and, by Bland's rule, $c_s^2 \leq 0$ since otherwise x_t would not be the entering variable in T_2 . Hence,

$$\sum_{i=1}^m a_{is}^1 c_{j_i}^2 < 0$$

implying that there exists k with $a_{ks}^1 c_{j_k}^2 < 0$. Notice that $k \neq r$, i.e. $j_k < t$, since the pivot element in T_1 , a_{rs}^1 , must be positive and $c_t^2 > 0$. However, in T_2 , all cost coefficients c_j^2 except c_t^2 are nonnegative; otherwise x_j would have been selected as entering variable. Thus $c_{j_k}^2 < 0$ and $a_{ks}^1 > 0$. This is a contradiction because Bland's rule should have selected x_{j_k} rather than x_t in T_1 as leaving variable. QED.

2.5 Phase I of the Simplex Method

In this section, we show how to transform a linear program into the form presented in Section 2.2. For that purpose, we show how to find a basis of the linear program which leads to a basic feasible solution. Sometimes, of course, we may inherit a bfs as part of the problem formulation. For example, we might have constraints of the form $Ax \leq b$ with $b \geq 0$ in which case the slack variables constitute a bfs. Otherwise, we use the two-phase simplex method to be described in this section.

Consider a linear program in canonical form with $b \geq 0$ (this latter restriction is without loss of generality since we may multiply some constraints by -1). In phase I,

instead of solving

$$\begin{aligned} & \text{Max } z = c_0 + c^T x \\ & \text{subject to:} \\ (P) \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

we add some artificial variables $\{x_i^a : i = 1, \dots, m\}$ and consider the linear program:

$$\begin{aligned} & \text{Min } w = \sum_{i=1}^m x_i^a \\ & \text{subject to:} \\ & Ax + Ix^a = b \\ & x \geq 0, x^a \geq 0. \end{aligned}$$

This program is not in the form required by the simplex method but can easily be transformed to it. Changing the $\min w$ by $\max w' = -w$ and expressing the objective function in terms of the initial variables, we obtain:

$$\begin{aligned} & \text{Max } w' = -e^T b + (e^T A)x \\ & \text{subject to:} \\ (Q) \quad & Ax + Ix^a = b \\ & x \geq 0, x^a \geq 0 \end{aligned}$$

where e is a vector of 1's. We have artificially created a bfs, namely $x = 0$ and $x^a = b$. We now use the simplex method as described in the previous section. There are three possible outcomes.

1. w' is reduced to zero and no artificial variables remain in the basis, i.e. we are left with a basis consisting only of original variables. In this case, we simply delete the columns corresponding to the artificial variables, replace the objective function by the objective function of (P) after having expressed it in terms of the nonbasic variables and use Phase II of the simplex method as described in Section 2.3.
2. $w' < 0$ at optimality. This means that the original LP (P) is infeasible. Indeed, if x is feasible in (P) then $(x, x^a = 0)$ is feasible in (Q) with value $w' = 0$.
3. w' is reduced to zero but some artificial variables remain in the basis. These artificial variables must be at zero level since, for this solution, $-w' = \sum_{i=1}^m x_i^a = 0$. Suppose

that the i th variable of the basis is artificial. We may pivot on any nonzero (not necessarily positive) element $\bar{a}_{ij}$ of row i corresponding to a non-artificial variable x_j . Since $\bar{b}_i = 0$, no change in the solution or in w' will result. We say that we are *driving the artificial variables out of the basis*. By repeating this for all artificial variables in the basis, we obtain a basis consisting only of original variables. We have thus reduced this case to case 1.

There is still one detail that needs consideration. We might be unsuccessful in driving one artificial variable out the basis if $\bar{a}_{ij} = 0$ for $j = 1, \dots, n$. However, this means that we have arrived at a zero row in the original matrix by performing elementary row operations, implying that the constraint is redundant. We can delete this constraint and continue in phase II with a basis of lower dimension.

Example

Consider the following example already expressed in tableau form.

$-z$	x_1	x_2	x_3	x_4	
	20	16	12	5	0
	1	0	1	2	4
	0	1	2	3	2
	0	1	0	2	2

We observe that we don't need to add three artificial variables since we can use x_1 as first basic variable. In phase I, we solve the linear program:

w	x_1	x_2	x_3	x_4	x_1^a	x_2^a	
		2	2	5			4
	1	0	1	2			4
		1	2	3	1		2
		1	0	2		1	2

The objective function is to minimize $x_1^a + x_2^a$ and, as a result, the objective function coefficients of the nonbasic variables as well as $-\bar{c}_0$ are obtained by taking the negative of the sum of all rows corresponding to artificial variables. Pivoting on $\bar{a}_{22}$, we obtain:

w	x_1	x_2	x_3	x_4	x_1^a	x_2^a	
			-2	-1	-2		0
	1		1	2	0		4
		1	2	3	1		2
			-2	-1	-1	1	0

This tableau is optimal and, since $w = 0$, the original linear program is feasible. To obtain a bfs, we need to drive x_1^a out of the basis. This can be done by pivoting on say $\bar{a}_{34}$. Doing so, we get:

w	x_1	x_2	x_3	x_4	x_1^a	x_2^a	
			0		-1	-1	0
	1		-3		-2	2	4
		1	-4		-2	3	2
			2	1	1	-1	0

Expressing z as a function of $\{x_1, x_2, x_4\}$, we have transformed our original LP into:

$-z$	x_1	x_2	x_3	x_4	
			126		-112
	1		-3		4
		1	-4		2
			2	1	0

This can be solved by phase II of the simplex method.

Problems

Problem 2.1. Solve by the simplex method:

Max $z = 10 + 2x_2 + 3x_5$
subject to:

$$x_1 - x_2 + x_5 = 4$$

$$3x_2 + x_3 - x_5 = 12$$

$$x_2 + x_4 + 2x_5 = 14$$

$$2x_2 + x_5 + x_6 = 13$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0.$$

Show all intermediate tableaux.

Problem 2.2. Solve by the simplex method using *only one pivot*:

$$\begin{aligned} \text{Max } z &= x_1 + 4x_2 + 5x_3 \\ \text{subject to:} \\ x_1 + 2x_2 + 3x_3 &\leq 2 \\ 3x_1 + x_2 + 2x_3 &\leq 2 \\ 2x_1 + 3x_2 + x_3 &\leq 4 \\ x_1 \geq 0, x_2 \geq 0, x_3 &\geq 0. \end{aligned}$$

Problem 2.3. Solve by the two-phase simplex method:

$$\begin{aligned} \text{Max } z &= 3x_1 + x_2 \\ \text{subject to:} \\ x_1 - x_2 &\leq -1 \\ -x_1 - x_2 &\leq -3 \\ 2x_1 + x_2 &\leq 4 \\ x_1 \geq 0, x_2 &\geq 0. \end{aligned}$$

Problem 2.4. Solve by the simplex method:

$$\begin{aligned} \text{Max } z &= x_{11} + 2x_{12} + 3x_{21} + 4x_{22} + 5x_{31} + 7x_{32} \\ \text{subject to:} \\ x_{11} + x_{12} &\leq 1 \\ x_{21} + x_{22} &\leq 1 \\ x_{31} + x_{32} &\leq 1 \\ x_{11} + x_{21} + x_{31} &\leq 1 \\ x_{12} + x_{22} + x_{32} &\leq 1 \\ x_{ij} &\geq 0 \quad i \in \{1, 2, 3\}, j \in \{1, 2\}. \end{aligned}$$

Were you expecting the optimum solution to have all components either 0 or 1?

Problem 2.5. Find a feasible solution to the following system:

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 + x_5 &= 2 \\ -x_1 + 2x_2 + x_3 - 3x_4 + x_5 &= 1 \\ x_1 - 3x_2 - 2x_3 + 2x_4 - 2x_5 &= -4 \\ x_1, x_2, x_3, x_4, x_5 &\geq 0 \end{aligned}$$

Problem 2.6. Use the simplex method to show that the following constraints imply $x_1 + 2x_2 \leq 8$:

$$4x_1 + x_2 \leq 4$$

$$2x_1 - 3x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

Problem 2.7. How are the various rules of the simplex method affected when solving a minimization problem instead of a maximization problem as described in these notes?